Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Planning and Soil Testing

1.1 Topic Name, Scope, and Agronomic Purpose

- **Definition and Agronomic Framework:** Pre-sowing crop planning and diagnostic soil testing for cotton (*Gossypium* spp.) establish systematic, science-based protocols for site selection, land evaluation, soil health assessment, cropping system design, and precision nutrient budgeting prior to field operations.
- **Core Agronomic Purpose:** Soil diagnostic testing identifies baseline chemical, physical, and biological constraints, allowing growers to correct macro- and micronutrient deficiencies, regulate soil pH and salinity, break hardpans, and optimize fertilizer application efficiency. Timely planning prevents early seedling mortality, protects crop stands against soil-borne pathogens and nematodes, and maximizes economic lint yields.
- **Prevention of Biotic and Abiotic Stresses:** Comprehensive pre-season planning helps farmers avoid cultivating susceptible cultivars in disease hotspots, prevents excessive nitrogenous fertilizer applications that attract sap-sucking pests, and establishes robust surface drainage infrastructure to prevent crop waterlogging and parawilt.
- **Precision Agriculture Integration:** Serves as the foundational step for site-specific nutrient management (SSNM) and Variable Rate Technology (VRT), ensuring balanced input management across diverse soil management zones.

1.2 Land Selection, Topography, and Site Suitability Criteria

- **Soil Type Suitability Parameters:** Cotton thrives across diverse soil textures, provided they offer adequate effective depth (> 1.0 meter), high water-retention capacity, and good internal drainage. Highly productive soil orders include:
  - *Deep Alluvial Soils (Inceptisols / Alfisols):* Dominant in the irrigated North Zone (Punjab, Haryana, Rajasthan); deep, fertile loams and clay loams.
  - *Medium to Heavy Black Soils (Vertisols):* Dominant in Central and South India (Maharashtra, Gujarat, MP, Telangana, Karnataka); high montmorillonite clay content, excellent moisture retention, self-mulching habit.
  - *Red Loamy and Lateritic Soils (Alfisols / Ultisols):* Common in rainfed southern tracts; require organic enrichment and frequent irrigation/rainfall.
- **Unsuitable Land Characteristics:**

  - *Very Coarse Sandy Soils:* Low cation exchange capacity (CEC), poor moisture retention, high nutrient leaching risk.
  - *Highly Saline / Sodic Soils:* Electrical conductivity (EC) > 4.0 dS/m or Exchangeable Sodium Percentage (ESP) $> 15\%$ causes osmotic stress, seedling scorching, and root stunting.
  - *Shallow Soils with Subsurface Hardpans:* Soil depth < 45 cm or dense pans at $20 - -35$ cm restricts taproot penetration, inducing right-angle root bending and drought susceptibility.
  - *Waterlogged Lowlands:* Fields subject to stagnation (> 24 hours) cause root hypoxia/anoxia, root rot, and parawilt.

- **Topography and Slope Management:** Select level to gently sloping land ($< 1 - -2\%$ slope). Uniform field topography facilitates precision laser land levelling, ensuring uniform irrigation distribution and preventing localized water pooling.

- **Irrigation Water Quality Assessment:** Evaluate proximity and reliability of canal water or tubewell infrastructure. High-salinity or high-RSC (Residual Sodium Carbonate) tubewell water must be tested before pre-sowing irrigation (*Rauni*).

1.3 Agro-Climatic Requirements and Environmental Planning

- **Thermal Regimes and Phenological Thresholds:**
  - *Germination Phase:* Minimum soil temperature required is 15°C; optimal seedbed temperature is 20°C to 30°C at 5 cm depth for rapid emergence (4–8 DAS).
  - *Vegetative Growth Phase:* Optimal day temperature range is 25°C to 35°C. Ambient temperatures $> 40 - -42$°C during emergence cause severe hypocotyl scorching and seedling burning.
  - *Reproductive Phase:* Optimal temperature for flowering and boll setting is 27°C to 32°C. High night temperatures (> 28°C) or extreme day heat (> 38°C) induce pollen sterility, square abortion, and boll shedding. Cold nights (< 15°C) cause leaf reddening and anthocyanin accumulation.

- **Sunshine and Moisture Requirements:**
  - *Solar Radiation:* Cotton is a sun-loving plant requiring abundant solar radiation. Prolonged cloudy weather during flowering increases square and young boll drop and favors fungal leaf spots.
  - *Water Requirement:* Consumes 600 to 800 mm of water throughout its growth cycle. Requires well-distributed rainfall or 4–6 planned irrigations.

- **Agro-Ecological Zonation Planning across India:**
  - *North Zone (Punjab, Haryana, Rajasthan):* Fully irrigated belt; cotton-wheat/mustard rotations; short window (April 1 to May 15) to escape Cotton Leaf Curl Virus (CLCuD) and pink bollworm.
  - *Central Zone (Gujarat, Maharashtra, MP):* Rainfed black soil belt; monsoon-dependent sowing (June–July); long crop duration (160–180 DAS).

  - *South Zone (Karnataka, AP, Telangana, Tamil Nadu):* Mixed rainfed/irrigated belt; Kharif and late-summer/winter cropping windows.

1.4 Physicochemical Soil Quality Indicators and Baseline Parameters

- **Soil Texture and Porosity:** Ideal physical tilth comprises a balanced mixture of sand, silt, and clay (loam to clay loam) with 50% total pore space (equally divided between macropores for aeration and micropores for water storage).
- **Soil Reaction (pH) Limits:**
  - *Optimal pH Range:* 6.0 to 8.0.
  - *Tolerance Limits:* Cotton tolerates moderately alkaline soils up to pH 8.5, provided internal drainage is unimpeded.
  - *Acidic Soils ($pH < 5.5$):* Induces Aluminum and Manganese toxicity, fixing applied Phosphorus; requires liming ($CaCO_3$) prior to land preparation.
- **Electrical Conductivity (EC) Thresholds:**
  - Baseline soil EC should be < 2.0 dS/m (in 1:2 soil-water extract).
  - Germinating seeds and young seedlings are sensitive to salinity; EC > 3.0 dS/m severely reduces germination percentage and seedling vigor.
- **Soil Organic Carbon (SOC) Benchmarks:**
  - Baseline minimum SOC threshold is $> 0.50\%$.
  - Target SOC for high-yielding soil health is $> 0.75\%$ to 1.0%. Adequate SOC improves cation exchange capacity, enhances beneficial microbial activity, and buffers soil structure against compaction.
- **Bulk Density and Subsurface Hardpan Diagnostics:**
  - Ideal topsoil bulk density ranges between 1.25 and 1.40 g/cm^3.
  - Bulk density > 1.55 g/cm^3 at $20 - -35$ cm depth indicates dense subsurface hardpans created by continuous shallow tillage or intensive puddled rice-wheat rotations, requiring subsoiling.

1.5 Standardized Soil Sampling Procedures and Field Grid Protocols

- **Sampling Schedule and Frequency:**
  - Collect soil samples during the pre-season off-period (February to April), immediately following the harvest of preceding Rabi crops and prior to applying farmyard manure, fertilizers, or pre-sowing *Rauni* irrigation.
  - Conduct full soil fertility testing once every 2 to 3 years for macronutrients, and every 3 to 4 years for micronutrients.

- **Field Sampling Grid and Pattern:**
  - Divide fields into homogeneous management zones based on soil color, texture, slope, crop history, and yield performance. Avoid mixing samples across distinct soil types.
  - Traverse each management zone in a 'Z-shaped' or 'W-shaped' zig-zag pattern, collecting 15 to 20 individual core sub-samples per 1–2 hectare field unit.
- **Sampling Depth Standards:**
  - *Standard Topsoil Depth (0–15 cm):* Evaluates Organic Carbon, pH, EC, available P, K, S, and micronutrients (Zn, Fe, Cu, Mn, B).
  - *Subsoil Profile Depth (15–30 cm or 30–60 cm):* Evaluates residual Nitrate-Nitrogen ($NO_3{-}N$), subsoil salinity, and hardpan chemical traits.
- **Sampling Tools and Excluded Sampling Zones:**
  - Use stainless steel tube augers, screw augers, or clean iron spades (using the V-shaped slice method, taking a 2.5 cm thick core from 0–15 cm depth).
  - *Strict Exclusions:* Never collect samples near field borders, bunds, irrigation channels, manure storage pits, tree shade zones, compost heaps, or fertilizer spill spots.
- **Specialized Nematode Soil Sampling Protocol:**
  - Collect samples when root-knot (*Meloidogyne incognita*) and reniform (*Rotylenchulus reniformis*) nematode populations are at their peak (immediately before or after harvest).
  - Take 20 to 22 core samples at 15–20 cm depth directly inside moist root-zone soil.
  - Mix cores gently in a plastic bucket, pack 1.0 kg soil into a sealed plastic bag (never paper bags, which dry out and kill nematodes), keep in shade, and submit immediately to a nematology lab.

1.6 Soil Sample Processing, Handling, and Laboratory Submission

- **Sample Homogenization and Quartering Method:**
  - Thoroughly mix the 15 to 20 collected sub-samples in a clean plastic bucket to form a composite bulk sample (2.0 to 3.0 kg).
  - Spread the composite soil uniformly on a clean plastic sheet, divide into four equal quarters, discard two diagonally opposite quarters, remix the remaining soil, and repeat this quartering process until a representative 500-gram sample remains.
- **Air-Drying Protocol:**
  - Air-dry the 500 g soil sample under indoor shade on clean polythene sheets or glass trays at room temperature ($25 - -30^{\circ}$C).

  - *Warning:* Never dry soil samples under direct sunlight, inside ovens, or near high heat sources, as high temperatures volatilize soil Nitrogen, alter microbial dynamics, and distort micronutrient availability.

- **Grinding, Sieving, and Packaging:**
  - Gently crush air-dried soil clods using a clean wooden mallet and mortar (avoiding metal hammers that break mineral grains).
  - Pass the pulverized soil through a 2.0 mm stainless steel sieve.
  - Pack the sieved soil in clean, moisture-proof polythene bags.

- **Labelling and Documentation Tagging:**
  - Attach two waterproof identification tags (one inside the bag, one tied outside).
  - Record essential sample details: Farmer name, village/district, field ID/survey number, GPS coordinates, sampling depth, preceding crop, proposed cultivar (Bt/Desi cotton), irrigation source, and specific testing requested.

- **Laboratory Submission:** Submit sealed samples to approved government soil testing laboratories, State Agricultural University (SAU) facilities, or NABL-accredited diagnostic labs.

1.7 Soil Testing Analytical Methods and Key Diagnostic Parameters

- **Physical and Physicochemical Diagnostic Methods:**
  - *Soil Reaction (pH):* Measured in a 1:2 soil-water suspension using a calibrated digital pH meter with a glass electrode.
  - *Electrical Conductivity (EC):* Measured in a 1:2 soil-water supernatant using a digital conductivity meter (expressed in dS/m or dS/m at 25°C).
  - *Soil Organic Carbon (SOC):* Analyzed via the Walkley-Black wet oxidation method using standard Potassium Dichromate ($K_2Cr_2O_7$) and concentrated Sulfuric Acid (H_2SO_4), back-titrated with Ferrous Ammonium Sulfate.

- **Primary Macronutrient Extraction Methods:**
  - *Available Nitrogen (N):* Determined via Subbiah and Asija's Alkaline Potassium Permanganate ($KMnO_4$) distillation method (expressed in kg N/ha).
  - *Available Phosphorus (P):* Extracted using Olsen's reagent ($0.5MNaHCO_3$ solution at pH 8.5) for neutral-to-alkaline soils, measured spectrophotometrically at 882 nm using ascorbic acid blue color development (expressed in kg P/ha or kg P_2O_5/ha).
  - *Available Potassium (K):* Extracted using neutral $1.0N$ Ammonium Acetate (NH_4OAc) at pH 7.0 and measured via Flame Photometry (expressed in kg K/ha or kg K_2O/ha).

- **Secondary and Micronutrient Extraction Methods:**

  - *Available Sulfur (S):* Extracted with $0.15\% CaCl_2$ solution and measured turbidimetrically using barium chloride ($BaCl_2$).
  - *DTPA-Extractable Micronutrients (Zn, Fe, Cu, Mn):* Extracted using $0.005M$ DTPA + $0.01M CaCl_2$ + $0.1M$ TEA buffer solution at pH 7.3 and quantified using an Atomic Absorption Spectrophotometer (AAS) or ICP-OES (expressed in mg/kg or ppm).
  - *Hot-Water Soluble Boron (B):* Extracted in boiling water and quantified using the Azomethine-H colorimetric method.

1.8 Interpretation of Soil Test Results and Target Benchmark Ranges

- **Soil Reaction (pH) and EC Interpretation Matrix:**
  - $pH < 6.0$ *(Acidic):* Low availability of P, Ca, Mg; requires agricultural lime ($CaCO_3$).
  - *pH 6.5–7.8 (Optimal):* Balanced availability of all essential nutrients.
  - $pH > 8.5$ *(Alkaline/Sodic):* Micronutrient deficiencies (Zn, Fe, B); high ESP; requires Gypsum ($CaSO_4 \cdot 2H_2O$) amendment.
  - $EC < 1.0$ *dS/m (Normal):* Safe for seed germination and seedling growth.
  - EC $1.0 - -2.0$ *dS/m (Critical):* Sensitive cotton varieties show slight growth reduction.
  - $EC > 2.0$ *dS/m (Saline):* Severe germination inhibition; requires heavy leaching with canal water.
- **Soil Fertility Rating Scale for Cotton Soils:**
  - *Soil Organic Carbon (SOC):* Low ($< 0.50\%$), Medium ($0.50 - -0.75\%$), High ($> 0.75\%$).
  - *Available Nitrogen:* Low (< 280 kg N/ha), Medium ($280 - -560$ kg N/ha), High (> 560 kg N/ha).
  - *Available Phosphorus (P_2O_5):* Low (< 22.5 kg P_2O_5/ha), Medium ($22.5 - -55.0$ kg P_2O_5/ha), High (> 55.0 kg P_2O_5/ha).
  - *Available Potassium (K_2O):* Low (< 140 kg K_2O/ha), Medium ($140 - -280$ kg K_2O/ha), High (> 280 kg K_2O/ha).
- **Critical Micronutrient Diagnostic Thresholds (DTPA Extractable):**
  - *Zinc (Zn):* Critical limit is 0.60 ppm (mg/kg). If < 0.60 ppm, apply Zinc Sulphate heptahydrate (21%) @ 10 kg/acre at sowing.
  - *Boron (B):* Critical limit is 0.50 ppm. If < 0.50 ppm, apply Borax @ 4.0 kg/acre (400 g elemental B) as a basal soil application.
  - *Iron (Fe):* Critical limit is 4.50 ppm.
  - *Manganese (Mn):* Critical limit is 2.00 ppm.
  - *Copper (Cu):* Critical limit is 0.20 ppm.

1.9 Soil Test-Based Fertilizer and Amendment Recommendations

- **Nutrient Uptake and Removal Benchmarks:** High-yielding cotton crops producing 25–30 q/ha seed cotton remove approximately 120–150 kg Nitrogen, 40–50 kg Phosphorus (P_2O_5), and 120–160 kg Potassium (K_2O) per hectare from the soil profile.
- **Fertilizer Adjustment Rules Based on Soil Test Ratings:**
  - *Low Soil Rating Category:* Increase state recommended fertilizer dose (RDF) by 25% for that specific nutrient.
  - *Medium Soil Rating Category:* Apply 100% of state recommended fertilizer dose (RDF).
  - *High Soil Rating Category:* Reduce state recommended fertilizer dose (RDF) by 25% for that specific nutrient.
- **State Agricultural University (SAU) Baseline Fertilizer Schedules (per acre):**
  - *Punjab Baseline (Bt Hybrids):* 80 kg Urea, 27 kg DAP (or 75 kg SSP). On light soils, add 20 kg Muriate of Potash (MOP).
  - *Haryana Baseline (Bt Hybrids):* 150 kg Urea, 50 kg DAP (or 150 kg SSP), 40 kg MOP.
  - *Rajasthan Baseline (Bt Hybrids):* 130 kg Urea, 35 kg DAP (or 100 kg SSP), 14 kg MOP.
- **Basal Soil Organic Enrichment and Amendments:**
  - *Farmyard Manure (FYM):* Broadcast 5 to 10 t/ha (or 5 t/acre) well-decomposed FYM or compost 2–3 weeks before sowing during secondary tillage.
  - *Neem Cake Enrichment:* Incorporate Neem Cake @ 250 kg/ha (5 q/ha) during field preparation to suppress subterranean termites and nematodes while retarding nitrification.
  - *Gypsum Application in Sodic Soils:* Apply Gypsum @ $2.5 - -5.0$ t/ha based on soil gypsum requirement (GR) tests to reduce ESP and improve water infiltration.
- **Bio-Fertilizer and Biological Soil Inoculation:**
  - Mix *Azotobacter / Azospirillum* (free-living/associative N-fixer) + Phosphate Solubilizing Bacteria (PSB) @ 10 kg/ha each with 200 kg moist FYM.
  - Mix *Trichoderma viride / T. harzianum* (1% WP @ 10 kg/ha) with 200 kg FYM, incubate for 7 days under shade, and broadcast during final seedbed preparation to suppress soil-borne fungal pathogens (*Rhizoctonia*, *Fusarium*).

1.10 Cropping System Planning, Rotation Schemes, and Refugium Design

- **Selection of Crop Rotation Schemes:** Continuous cotton-cotton monoculture increases soil-borne fungal pathogens (*Fusarium*, *Rhizoctonia*), expands nematode populations (*Meloidogyne*), depletes subsoil nutrients, and encourages mealybug buildup. Rotate cotton with non-host crops at least once every 2 to 3 years:
  - *Cotton – Wheat Rotation:* Dominant 1-year rotation in North India. Requires short-duration cotton cultivars to allow timely wheat sowing by November 15.
  - *Cotton – Mustard (Raya) Rotation:* Water-saving 1-year rotation suitable for limited irrigation conditions.
  - *Cotton – Chickpea (Bengal Gram) Rotation:* Pulses restore soil Nitrogen reserves and improve soil organic matter.
  - *Cotton – Barley / Berseem (Fodder) Rotations:* Excellent for soil health restoration.
  - *Cotton – Sunflower Rotation:* Alternative oilseed rotation.
- **Prohibition of Host Crop Proximity:**
  - Avoid intercropping or growing alternate host crops near cotton fields.
  - *High-Risk Alternate Hosts:* Okra (*Abelmoschus esculentus*), cucurbits, brinjal, tomato, castor, pigeonpea, and *Dhaincha*. These crops harbor early-season whiteflies, thrips, and aphids, and serve as virus inoculum reservoirs for Cotton Leaf Curl Virus (CLCuD) and Tobacco Streak Virus (TSV).
  - Avoid intercropping green gram (Moong) inside cotton in whitefly-prone endemic zones.
- **Refugia Strategy Planning for Bt Cotton (Bollgard II):**
  - To delay pink bollworm (*Pectinophora gossypiella*) and American bollworm (*Helicoverpa armigera*) resistance against Cry1Ac and Cry2Ab endotoxins, refugium planning is mandatory.
  - *Structured Refugia Layout:* Plant 5 border rows of non-Bt cotton around the outer perimeter of the Bt cotton field (or allocate 20% field area under non-Bt cotton).
  - *Refugia-In-Bag (RIB):* Use commercial RIB seed bags containing $5 - -10\%$ non-Bt seeds pre-blended inside the Bt seed bag.

1.11 Water Resources, Irrigation Quality, and Drainage Planning

- **Source Water Quality Diagnostics:** Test canal water and tubewell water samples quarterly in accredited laboratories before using for pre-sowing *Rauni* irrigation or seasonal flooding.
- **Irrigation Water Quality Classification Standards:**

  - *Electrical Conductivity (ECw):* Good quality (< 1.5 dS/m); Marginal quality ($1.5 - -3.0$ dS/m); Unsuitable quality (> 3.0 dS/m).
  - *Sodium Adsorption Ratio (SAR):* $SAR = \frac{Na^+}{\sqrt{\frac{Ca^{2+}+Mg^{2+}}{2}}}$. Safe ($< 10$); Sodic hazard ($10 - -18$); Severe sodic hazard ($> 18$).
  - *Residual Sodium Carbonate (RSC):* $RSC = (CO_3^{2-} + HCO_3^-) - (Ca^{2+} + Mg^{2+})$ in meq/L. Safe (< 1.25 meq/L); Marginal ($1.25 - -2.50$ meq/L); Unsafe (> 2.50 meq/L).

- **Management of Saline / High-RSC Tubewell Water:**
  - Always apply pre-sowing *Rauni* irrigation using good-quality canal water. Never use poor-quality tubewell water for pre-sowing *Rauni*.
  - Alternate tubewell water with canal water irrigations.
  - Treat high-RSC water by passing it through Gypsum beds in irrigation channels before field application.

- **Surface Drainage Infrastructure Planning:**
  - Cotton roots are sensitive to water stagnation (> 24 hours causes root hypoxia and wilting).
  - Construct deep peripheral drainage channels along field boundaries connected to main regional drainage outlets.
  - Level fields using Laser Land Levellers prior to bed formation to eliminate low spots where rainwater accumulates.

1.12 Integrated Step-by-Step Pre-Season Planning and Soil Testing Roadmap

- **Chronological Pre-Season Action Roadmap:**
  1. *Off-Season Diagnostic Phase (February 1 to March 15):*
     - Inspect field history for disease, weed, and nematode hotspots.
     - Collect 15–20 soil cores (0–15 cm and 15–30 cm depth) per management zone following the zig-zag pattern.
     - Collect specialized root-zone soil samples for nematode testing.
     - Air-dry, quarter, package, label, and submit samples to accredited soil testing laboratories.
     - Test tubewell irrigation water quality for EC, SAR, and RSC.
  2. *Planning Input Procurement Phase (March 15 to April 1):*
     - Receive soil test diagnostic report; evaluate pH, EC, SOC, available NPK, and micronutrient statuses (Zn, B, Fe).
     - Calculate adjusted fertilizer requirements based on soil test rating categories.
     - Select approved, zone-recommended CLCuD-resistant Bt hybrids or Desi cotton varieties from authorized dealers; verify color-coded certification tags.

- Procure required basal fertilizers (DAP/SSP, MOP, Urea, $ZnSO_4$), organic inputs (FYM, Neem cake), and bio-agents (*Trichoderma*, PSB, *Azotobacter*).
- Design crop rotation schedule and refugia planting strategy.

3. *Primary Field Preparation Land Amendment Phase (April 1 to April 15):*
   - Execute deep summer Mouldboard ploughing to invert topsoil and expose soil-borne pests/pathogens to solar heat.
   - Cross-subsoil with a chisel plough at 1.0 m spacing to break subsurface hardpans.
   - Broadcast basal organic inputs: 5–10 t/ha FYM, 250 kg/ha Neem cake, and *Trichoderma*-enriched FYM (10 kg/ha in 200 kg FYM). Apply Gypsum if sodic soil amendments are indicated by soil test.
   - Level field using a Laser Land Leveller to achieve uniform topography.
4. *Pre-Sowing Irrigation Final Seedbed Phase (April 15 to May 15):*
   - Construct surface drainage ditches along field borders.
   - Apply heavy pre-sowing *Rauni* irrigation using canal water to charge the soil profile.
   - Wait for soil to reach *Wattar* (workable moisture) condition; cultivate 2 times with tillers and plank with *Suhaga* to seal soil moisture.
   - Execute acid delinting (H_2SO_4 @ 100 mL/kg) and multi-tier seed dressings ($F \rightarrow I \rightarrow B$) prior to sowing.
   - Drill all basal Phosphorus, Potassium, and Zinc Sulphate at sowing based on soil test recommendations.

- **Core Planning Soil Testing Operational Integration Summary:**
  - *Baseline Soil Targets:* Deep loam/clay loam soil; pH 6.0–8.0; EC < 2.0 dS/m; Organic Carbon $> 0.75\%$; Bulk density < 1.4 g/cm^3.
  - *Sampling Protocol:* 15–20 sub-samples per zone at 0–15 cm and 15–30 cm depth in Feb–March; shade drying; quartering to 500 g; submission to accredited lab.
  - *Diagnostic Thresholds:* Low rating (+25% RDF), Medium rating (100% RDF), High rating (-25% RDF). Zinc < 0.6 ppm requires 10 kg/acre $ZnSO_4$ (21%); Boron < 0.5 ppm requires 4 kg/acre Borax.
  - *Field Execution Integration:* Laser levelling; MB ploughing + subsoiling at 1.0 m spacing; 5–10 t/ha FYM + 250 kg/ha Neem cake + *Trichoderma* (10 kg/ha); canal water *Rauni* irrigation; non-Bt refugia (5 border rows or RIB 5–10%); non-host Rabi crop rotation (Wheat, Mustard, Chickpea).